



SCHOOL OF COMPUTER TECHNOLOGY

AASD 4014 Deep Learning I

Final Project

Presentation date: Last 2 class of the last week, Wed and/or Fri (will be a draw)

Final submission due date: One week after the last class (Friday of week after last week, end of the day)

Project: Using or building AI models.

- Students have 2 options to design and develop AI models or applications.

Project requirements.

Projects **can not** be based on **image tasks**, those will be covered in DL2

Option 1

- **Train, build and evaluate a deep learning model.**
 - Options include training one or more of the following.
 - Predictive models
 - Classification models
 - Sentiment models
 - NLP models
 - **Model types**
 - RNN, CNNs, Transformers, Seq2Seq

Few examples:

- 1) Predict the price for this house
- 2) Detect fraud activity (classification)
- 3) Predict the price of oil for the next year
- 4) Predict revenue per day
- 5) Classify text (NLP)
- 6) Sentiment analysis
- 7) ...

Option 2

- **Implement LLM solutions**
 - Options include training, fine tuning, transfer learning, a combination of
 - Chatbots
 - RAGs
 - Agents
 - **Important note**

- While training, transfer learning or finetuning may be optional. Your project **must be comprehensive** if you choose not to use training or transfer learning.
- **Example:** You want to create a multi-agent system; it would not be enough to have 3 agents communicating with each other, you must demonstrate solving **real world problems**.

General Requirements

- Projects must have some elements of EDA.
 - Class balance, correlation matrices etc.
 - Part of being a data scientist is being able to communicate important information through visualizations and summary statistics.

Group, Roles and Responsibilities

- **Groups should be formed within the first week of class** (Preferably on or before Wednesday)
- **Minimum group size:** 4
- **Max group size:** 8
- Groups **must** select a project manager; this project manager should be **clearly identified** in the report and presentation.
- The project manager should coordinate and manage the project, ensure deadlines are met, run meetings and take minutes of meeting.
- Ideally groups should meet 3 times per week.

Database:

No specific requirement on the database, of-course it should have at least 2 categories (for classification cases).

You are free to choose your dataset and decide what you want to work on. If there are groups that like to work on other types of data than text or time series, please talk to me, to make sure it is possible before starting the work.

Mainly the dataset for this project can include:

1. Time series (example price of a product over time)
2. Text (e.g. product reviews)

Report

- **You are required to use the template provided.**
 - The sections in the template are the **minimum** requirements
 - You may add more sections.

Report Sections

- Background and problem statement,

- Plan of attack, what is your approach toward this?
- The Dataset,
- The model you picked to solve the problem,
- Results, the model performance (test, valid), the loss, predictions... (like use of confusion matrices etc...)
 - Also include the steps you took to tune your hyperparameter, architecture etc... (tell the whole story along the way)
 - A. Did you have a benchmark model?
 - B. Did your plan of attack make sense, now you have a better idea?
 - C. Do you do differently next time?
- Conclusions

In your results, please comment and discuss the followings:

1. Basic data analysis, e.g. data imbalance, outliers, how to deal with them?
2. Evaluate the model, how?
3. Tuning your models, for example how your model changes when the model structure, architecture, hyperparameters are changed?
4. Are any of your models overfitted? Why?

What will be in your final submission:

1. Your report
 - Background and problem statement,
 - Plan of attack, what is your approach toward this problem?
 - The dataset (investigate the dataset, present findings),
 - The model you picked to solve the problem,
 - Results, the model performance (test, valid), the loss, predictions... (like the use of confusion matrices etc...)
 - Also include the steps you took to tune your hyperparameter, architecture etc... (tell the whole story along the way)
 - a. Did your plan of attack make sense, now you have a better idea?
 - a. Do you do differently next time?
 - Conclusions
2. All your code
3. Example input data with model predictions on them (only few)

Important: at the beginning of your report there should be a table indicating individual tasks (i.e. who did what?) this needs to be precise with enough information.

Presentation:

- Note that you must present your **progress/results** on the date specified.
- If your project is **incomplete** by the **presentation date**, you can report progress and report results in the report.
- The presentation of results should still be a **complete presentation**.

- Note that this date cannot be moved, the department needs the grade by end of the week.
- The presentation is a **15-minute** presentation with **5 minutes** allocated for questions.
- Being significantly over or under the 15-minute mark may result in lost marks. (14 mins or 16 mins is fine, 10 mins or 20 minutes is not)

(During presentation you don't need to have everything ready, you present what you have achieved until then)

There will be a draw on whom to present on which day.

Important Note:

1. Make sure to document your findings and back up your conclusions using the results, use visualization (this is important).
2. Only **one person** from each group to submit on BB (report to include all group members),
3. Your submission needs to include: your **data, code, report and your slides**.
4. If we test them, your codes should run without error.
5. **Important:** Compress all your files into zip (or similar) files and only submit that single file.
6. You can submit a few times, but only the latest submission is considered.

Evaluation:

- You will be marked on **two different components, individual** contribution and **group work**. Both components will be considered in
 1. The actual work
 2. Report
 3. Presentation
 4. Teamwork, communications and other
- Team will divide the work in a reasonable way, work will commence and will be monitored by managers but also everyone on the team.
- It is completely fine that team members do the same work, using different approaches or techniques (e.x. two or more people may train a model but with different architecture or parameters, another example: several people may contribute in writing a conclusion in the report), However this needs to be clearly identified in the report (under different sections, who did what). A table at beginning of the report need to identify who did what,
- Late submissions are subject to penalties (5 marks), but also depend on how long the delay is
- Any missing parts

Work will be marked on these items:

- Your general quality of the work
- Your approach and methodologies
- Your results/conclusions
- Your general quality of the report you submitted
- Your presentation quality and skills, (smoothness, clarity, slides, being on time and flow:

e.g.: Background and Problem>>Data>>task and responsibilities>>methodology>>results>>conclusion
(I understand you may not have the final results by time of submission)

Rubric

Criterion	D	C	B	A	A+
Comprehensive Project Design and Implementation	- The project lacks a clear problem statement or objective. - The chosen model type is not appropriate for the problem. - There is no evidence of group roles or responsibilities being defined.	- The project includes a basic problem statement and objective. - A model type is chosen but not well-justified. - Group roles are defined but not clearly documented in the report.	- The project includes a clear problem statement and objective. - The chosen model type is appropriate and justified. - Group roles and responsibilities are clearly documented in the report.	- The project includes a detailed problem statement and objective. - The chosen model type is appropriate, justified, and innovative. - Group roles and responsibilities are clearly documented and reflect effective collaboration	- The project includes a detailed problem statement and objective with real-world relevance. - The chosen model type is appropriate, justified, innovative, and effectively implemented. - Group roles and responsibilities are clearly documented, reflect effective collaboration, and demonstrate leadership and initiative.
Effective Data Analysis and Model Evaluation	- There is no evidence of exploratory data analysis (EDA). - Model evaluation metrics are missing or incorrect. - No attempt to address data	- Basic EDA is present but lacks depth (e.g., only class balance is considered). - Model evaluation metrics are present but not well-explained.	- EDA includes class balance and correlation matrices. - Model evaluation metrics are present and explained. - Some attempt to address data imbalance and outliers.	- EDA includes class balance, correlation matrices, and additional relevant analyses. - Model evaluation metrics are comprehensive and well-explained.	- EDA includes class balance, correlation matrices, and insightful analyses that inform model design. - Model evaluation metrics are comprehensive, well-explained, and include

	imbalance or outliers.	- Minimal attempt to address data imbalance or outliers.		Effective strategies to address data imbalance and outliers are implemented.	advanced techniques. - Innovative strategies to address data imbalance and outliers are implemented and justified.
Quality of Report and Presentation	- The report is missing key sections or lacks coherence. - The presentation is significantly under or over the time limit. - Visualizations are missing or unclear.	- The report includes all required sections but lacks detail. - The presentation is within the time limit but lacks engagement. - Visualizations are present but not well-integrated into the report.	- The report includes all required sections with sufficient detail. - The presentation is within the time limit and engages the audience. - Visualizations are clear and integrated into the report.	- The report includes all required sections with detailed analysis and insights. - The presentation is within the time limit, engaging, and well-structured. - Visualizations are clear, integrated, and enhance understanding.	- The report includes all required sections with detailed analysis, insights, and reflections. - The presentation is within the time limit, highly engaging, and professionally delivered. - Visualizations are clear, integrated, enhance understanding, and demonstrate creativity.